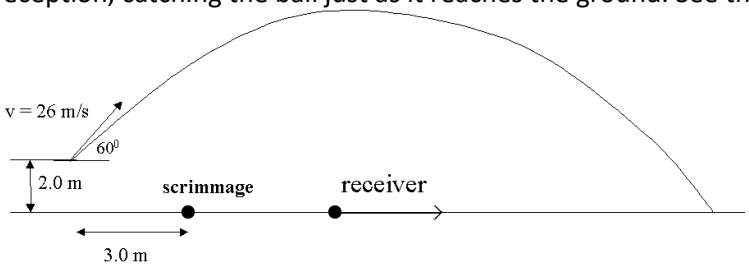
Problem Solving Review Questions

1. A football quarterback attempts a pass to one of the receivers. As the ball is snapped, the receiver leaves the line of scrimmage and runs directly down field. The quarterback releases the ball 2.0 s later and from a position 3.0 m behind the line of scrimmage. He throws the ball with a speed of 26 m/s at an elevation of 60° above the horizontal. The receiver makes a diving reception, catching the ball just as it reaches the ground. See the diagram below.



   a. What is the time of flight of the football? [4.7 s]
   b. What is the average speed of the receiver? [8.7 m/s]

2. An arrow is shot vertically upward with an initial speed of 25 m/s. When it's exactly halfway to the top of its flight, a second arrow is launched vertically upward from the same spot. The second arrow reaches the first arrow just as the first arrow reaches its highest point.
   a. What is the launch speed of the second arrow? [26.5 m/s]
   b. What maximum height does the second arrow reach? [35.8 m]

3. A truck travels at a constant speed of 28.0 m/s in the fast lane of a two-lane highway. It approaches a stationary car stopped at the side of the road. When the truck is still 1.2×10^2 m behind the car, the car pulls out into the slow lane with an acceleration of 2.6 m/s^2.
   a. How long will it take the truck to pass the car? [5.9 s]
   b. How far will the car have travelled when the truck passes it? [1.6×10^2 m]
   c. If the car were to maintain this acceleration, what would be its speed when it overtakes the truck? [41 m/s]

4. A clock has a second hand 15 cm long, from centre to tip. (Hint: the velocity of the tip of the second hand is always tangent to the circle)
   a. Draw a labelled diagram to represent this situation
   b. Calculate the instantaneous speed of the tip of the second hand. [1.57 cm/s]
   c. What is the instantaneous velocity of the tip at 15 s? [1.57 cm/s [down]]
   d. What is its change in velocity between 30 s and 45 s? [2.2 cm/s [up 45° right]]
   e. What is its average acceleration during the same time interval? (1 mark) [0.15 cm/s^2 [up 45° right]]

Textbook Questions:

Problems: page 25 #3, p 29 #3, 6, 10, 11, p 35 #10, 11, 12, p 58 #71, 72, 74, p 43 #1-3, 9, p 49 #1, 3, 5, 7,

Multiple Choice, True/False & Short Answer: p 53 #6, 7, 10-15, p 54 #2, 5, 6, 11, 13, 14, p 58#19, 20, 22, 27, 31, 32, 33, 34

work on following pages

Concepts in Kinematics

1. How many people are there in this room? N How many significant digits is this? ∞

   N = count; exact

2. Explain the difference between the following symbols:

   a. v_{av}, Δv, v_i

   $v_{av} = \frac{\Delta d}{\Delta t}$: average velocity

   $\Delta v = v_f - v_i$: change in velocity

   v_i : initial velocity

   b. $\vec{d}$, $\Delta\vec{d}$

   d : position from an origin

   Δd : displacement (change in position)

3. Two toddlers are 10 m apart and are moving towards each other, one travelling at 0.4 m/s and the other at 0.25 m/s. The faster child started 2.2 s after the slower one. Which of the following statements are true at the time the two children meet? (s = slow; f = fast)

   a. $\Delta t_f = \Delta t_s + 2.2s$

   b. $\Delta t_s = \Delta t_f + 2.2s$.

   Explain your answer: Slow child starts first, so walks 2.2 s longer.

4. The following vectors show the velocity of a roller coaster at two different points in time. What is the direction of the acceleration? Explain.

   $\vec{v}_1$ $\vec{v}_2$

   a is down and right, along $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$.

5. The velocity of a dragonfly at two different times is shown by the vectors below. Draw a vector diagram that would allow you to calculate the dragonfly's change in velocity.

   $\vec{v}_f$ $\vec{v}_i$

   $\Delta\vec{v} = \vec{v}_f + (-\vec{v}_i)$

6. The position of a student at two points in time is shown by the vectors below. Draw a tip-to-tail diagram that would show $\vec{d}_f - \vec{d}_i$. What does this vector represent physically?

   $\vec{d}_f$ $\vec{d}_i$ O

   $\Delta\vec{d} = \vec{d}_f + (-\vec{d}_i)$

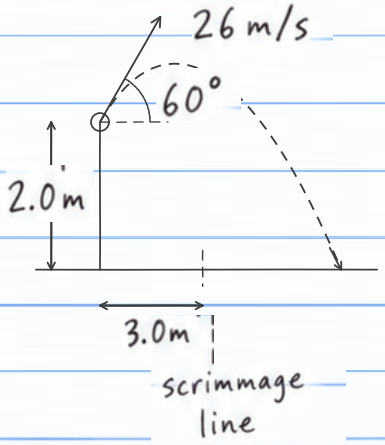
   displacement, from initial to final position

Review Questions

1) +y is up, +x is downfield

a) $v_{ix} = 26 \cos 60° = 13.0 \text{ m/s}$

$v_{iy} = 26 \sin 60° = 22.52 \text{ m/s}$



$\Delta d_y = -2.0 \text{ m}, \quad a_y = -9.81 \text{ m/s}^2$

$\Delta d_y = v_{iy} \Delta t + \frac{1}{2} a_y \Delta t^2$

$-2.0 = 22.52 \Delta t - 4.905 \Delta t^2$

$4.905 \Delta t^2 - 22.52 \Delta t - 2.0 = 0$

use quadratic formula

$\Delta t = -0.087 \text{ s}$ or 4.678 s

negative time is before the throw

$\Delta t_{ball} = 4.7 \text{ s}$

b) $\Delta d_{ball,x} = 13.0(4.6777) = 60.81 \text{ m}$

receiver starts 3.0 m ahead of the quarterback

$\Delta d_{receiver} = 60.81 - 3.0 = 57.81 \text{ m}$

$\Delta t_{receiver} = 4.6777 + 2.0 = 6.6777 \text{ s}$

$v_{av} = \frac{57.81}{6.6777}$

$v_{av} = 8.7 \text{ m/s}$

2) + is up

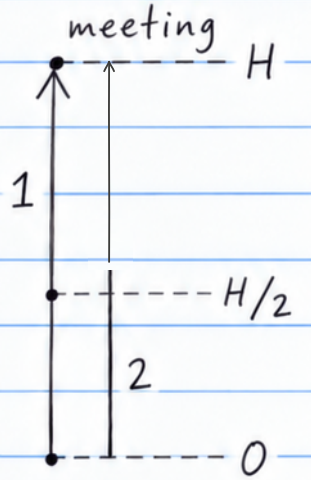
$\vec{v}_i = 25\ m/s$

at the top, $\vec{v}_f = 0$

$0 = 25^2 - 2(9.81)H$

$H = 31.86\ m$

$\Delta t_{top} = \frac{25}{9.81} = 2.548\ s$



halfway up:

$\Delta d = \frac{H}{2} = 15.93\ m$

$v_{half}^2 = 25^2 - 2(9.81)(15.93)$

$v_{half} = 17.68\ m/s$

$\Delta t_{half} = \frac{17.68 - 25}{-9.81} = 0.746\ s$

$2.548 - 0.746 = 1.802\ s$

a) $H = v_{i2}\Delta t - \frac{1}{2} g \Delta t^2$

$31.86 = v_{i2}(1.802) - 4.905(1.802)^2$

$v_{i2} = 26.5\ m/s\ [up]$

b) $0 = v_{i2}^2 - 2gH_2$

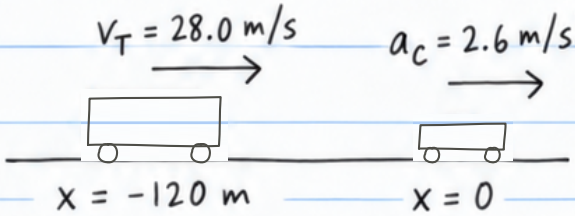
$H_2 = \frac{26.5165^2}{2(9.81)}$

$H_2 = 35.8\ m$

3) + is forward, car starts at $x = 0$

truck: $x_T = -120 + 28.0\,\Delta t$

car: $x_C = \frac{1}{2}(2.6)\,\Delta t^2$



same position when they pass

$1.3\,\Delta t^2 = -120 + 28.0\,\Delta t$

$1.3\,\Delta t^2 - 28.0\,\Delta t + 120 = 0$

use quadratic formula

$$\Delta t = \frac{28.0 \pm \sqrt{28.0^2 - 4(1.3)(120)}}{2.6}$$

$\Delta t = 5.904$ s or 15.634 s

a) truck passes first

$\Delta t = 5.9$ s

b) $\Delta d_C = \frac{1}{2}(2.6)(5.904)^2$

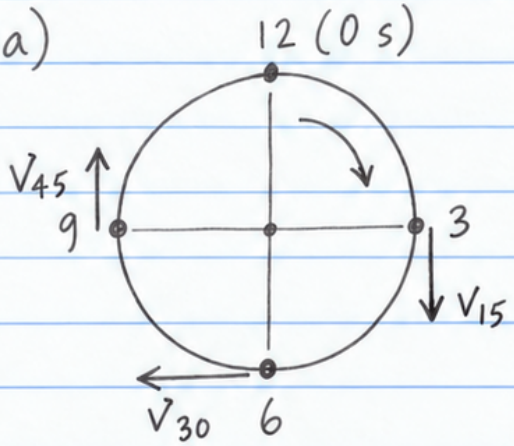
$\Delta d_C = 45$ m

c) use the later time when the car overtakes

$v_C = 0 + 2.6(15.634)$

$v_C = 41$ m/s

4a)



$r = 15$ cm

$T = 60$ s

b) $v = \dfrac{2\pi r}{T} = \dfrac{2\pi(15)}{60}$

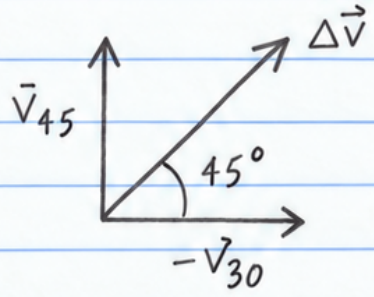
$v = 1.57$ cm/s

c) tangent at 15 s points down

$\vec{V}_{15} = 1.57$ cm/s [down]

d) $\Delta\vec{V} = \vec{V}_{45} + (-\vec{V}_{30})$

V_{30} is left, so $-V_{30}$ is right



$|\Delta V| = \sqrt{1.57^2 + 1.57^2} = 2.22$ cm/s

$\theta = \tan^{-1}\left(\dfrac{1.57}{1.57}\right) = 45°$

$\Delta V = 2.2$ cm/s [up 45° right]

e) $a_{av} = \dfrac{\Delta V}{\Delta t} = \dfrac{2.22}{15}$

$a_{av} = 0.15$ cm/s^2 [up 45° right]